PERFORMANCE ANALYSIS OF GREENBLATT'S MAGIC FORMULA IN THE BRAZILIAN MARKET

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NOTE ON THIS EDITION

This is a revised English edition, from November 2026, of the article written in Portuguese in 2025. While the code was being prepared for publication, a review found errors that affect the results. The tables, the charts and the numbers in the text below come from the corrected code, published with this article, so they differ from those of the Portuguese version, which is kept unchanged in this repository.

Corrections to the code and the data:

- Financial statements published on non-trading days were discarded, which left companies without indicators for a year or more. They now take effect on the next trading day.
- The sector filter missed the data provider's label "Financeiro e Outros" (Financials and Others), which covers 6 banks and 2 financial services companies. They are now excluded.
- The first rebalance of the quintile and Long-Short portfolios used the signal of the last date of the sample, a look-ahead error. It now uses the signal of the previous day.
- The three signals of the backtest were rebuilt with the corrected code, from the latest data collection.

Corrections to the article:

- The Ibovespa in Tables 1, 8 and 9 now covers the same period as the portfolios, without January 2, 2009, when they were not yet invested: 7.0% per year instead of 7.5%.
- Table 3 now counts the stocks of the final ranking; the previous values matched the ROC ranking.
- The Total row of Table 4 now shows the cumulative return; it used to show the final multiple (626.9% meant 6.27 times). The 30% Long-Short column became the 25% one, now the best.
- The note of Table 10 now gives the Sharpe formula of the Long-Short portfolios.
- Sections 3.2.1 and 3.3.3 now describe what the code does: Return on Capital uses the capital of four quarters earlier, and the factor regressions use monthly returns.
- Two statements about the literature were corrected: the sample of Fama and French (1992) and the year in which Buffett took control of Berkshire Hathaway.

What changed in the conclusions:

- The best Long-Short portfolio is the 25% one (CAGR of 3.3%, Sharpe of 0.21), not the 30% one, which fell from a CAGR of 4.1% and a Sharpe of 0.29 to 2.5% and 0.17.
- The alphas of the Magic Formula and of the Long-Short portfolios are not statistically significant, so the article no longer describes them as abnormal returns or a significant premium. Only the Magic Formula with Net Debt/Ebit and the 1st Quintile with Momentum have alphas significant at the 5% level.
- The Magic Formula is no longer described as consistently positive (6 negative years in 16) or as lower risk without qualification: its volatility is lower than that of the Ibovespa, but its maximum drawdown is deeper.

The central result remains: the Magic Formula portfolio returned 12.0% per year (12.2% in the 2025 version), above the Ibovespa and below the first quintile (13.9% in both versions).

1 INTRODUCTION

The Magic Formula is an investment strategy developed by Joel Greenblatt and presented in his book "*The Little Book That Beats the Market*" (2006). The methodology combines concepts of fundamental analysis, an approach widely used in equity investing that consists of assessing the quality of a company's financial indicators and business model to estimate its intrinsic value and identify buying opportunities. This perspective is part of the Value Investing philosophy, which seeks to buy stocks trading below their intrinsic value, exploiting pricing inefficiencies made by market participants.

Value Investing was created by Benjamin Graham and David Dodd in the book "*Security Analysis*" (1934) and consolidated by Graham in "The Intelligent Investor" (1949), becoming one of the most influential investment philosophies. Warren Buffett, a disciple of Graham, expanded the concept when he realized that the simple difference between market

price and intrinsic value was not enough to maximize long-term returns. His approach came to prioritize companies that are not only undervalued but also have sustainable competitive advantages over their competitors.

Graham's strategy seeks to generate excess returns consistently over the long term, exploiting persistent inefficiencies in the capital market. This view contrasts with the Efficient Market Hypothesis (EMH), formulated by Fama (1970), which is still widely discussed and debated today. According to the EMH, it is unfeasible to obtain abnormal returns systematically over time, since asset prices would fully reflect all available information. In this context, the rational investor should simply replicate the market portfolio, represented in Brazil by the Ibovespa index, rather than try to beat it.

Historically, few investors such as Buffett, Peter Lynch, Jim Simons and Joel Greenblatt have managed to beat their benchmarks consistently over long periods. Among them, Greenblatt stood out for distilling his strategy into a method that is easy to understand and implement.

This study seeks to replicate Greenblatt's strategy in the Brazilian context, building a theoretical portfolio for the period from 2009 to 2024. The performance obtained will be compared with that of the Ibovespa and assessed through complementary metrics, in order to verify whether active management is able to generate alpha relative to the reference index. The analysis covers annual performance, the division of the stock universe into quintiles and the formation of long-short portfolios, all based on the ranking defined by the Magic Formula.

2 THEORETICAL FRAMEWORK

This chapter aims to deepen the concepts and tools essential to understanding and evaluating the results obtained. First, we discuss previous studies that replicated Greenblatt's strategy both in the Brazilian market and in other international markets, allowing a comparative analysis with this work. Next, we explore topics fundamental to interpreting the strategy and its performance, including: the Efficient Market Hypothesis, Value Investing, Modern Portfolio Theory, risk factors and risk indicators.

2.1 Previous studies

Several researchers have tested Greenblatt's strategy in different geographies, generally with the aim of replicating the methodology, confronting its results with the efficient market hypothesis and checking whether applying the formula entails additional risk. The purpose of this section is to present the contributions of each study to the discussion of the topic.

In the Nordic market, Persson and Selander (2009) achieved satisfactory results: both portfolios built in the backtest beat the MSCI Nordic, showing that it is possible to exploit market inefficiencies in the long term. However, they did not find any significant advantage in terms of risk-adjusted return. Olin (2011) tested the Magic Formula in the Finnish market, varying portfolio size and rebalancing frequency. The study also built portfolios with the best and worst stocks in the ranking and compared the results with the OMX Helsinki index. The results went against the initial expectation: portfolios with quarterly rebalancing consistently beat longer horizons, regardless of the number of assets.

In Brazil, Zeidler (2014) and Milane (2016) applied the Magic Formula over similar periods (2000–2015) and obtained results consistent with those of the Finnish market, suggesting that the strategy can be replicated in different geographic contexts. Gunnar (2016) also applied the strategy in the Brazilian stock market, testing the Magic Formula and isolating

its two components — Earnings Yield and Return on Capital (ROC) — in order to assess the individual impact of each metric on the final return of the portfolios. The results showed that all portfolios beat the Ibovespa and the IbrX100; in particular, those based exclusively on Earnings Yield showed superior performance and lower volatility, while portfolios based only on ROC recorded lower returns and higher volatility.

Overall, the literature reviewed indicates that more concentrated portfolios, with 5 to 10 assets, tend to generate the highest returns, even after adjusting for risk. Even so, Greenblatt recommends more diversified portfolios, between 20 and 30 assets, to mitigate specific risks.

2.2 Efficient market hypothesis

Fama (1970) introduced the Efficient Market Hypothesis (EMH), one of the most influential concepts in modern financial theory, which inspired countless subsequent studies, both to validate it and to challenge it. The central premise of the EMH is that, if markets are truly efficient, it becomes extremely difficult to generate returns consistently above the market through analysis or trading strategies. In this scenario, the most recommended approach would be to build a broadly diversified portfolio that reflects the performance of the market as a whole.

The EMH has three fundamental premises: (i) market agents make rational decisions based on the available information, (ii) the prices of financial assets incorporate all past and present information and investors' future expectations, and (iii) there are no transaction costs to obtain information.

These premises sparked intense debates in academic and professional circles, leading to the development of several studies aimed at challenging the EMH. Critics of this theory seek to identify moments of market irrationality to support their arguments, such as the emergence of bubbles and abrupt drops in asset prices. Jegadeesh and Titman (1993) examined the Momentum effect, buying stocks that showed extraordinary returns over the last 6 to 12 months. Their results showed that these portfolios kept beating the market for up to one year after formation.

The size effect is also a widely recognized argument against the EMH. It suggests that companies with lower market value tend to generate higher returns than larger companies. This difference is attributed, in part, to the lower dissemination of information about smaller companies. Keim (1983) analyzed the annual return of small and large companies in the US market and observed that in half of the cases the additional return is due to seasonality in January, with 11% of these cases specifically related to the first week of the month.

2.3 Value Investing

Like Fama's (1970) Efficient Market Hypothesis, Value Investing has established itself as one of the most influential investment philosophies, based on the search for stocks of solid companies trading at prices significantly below their intrinsic value. For Graham and Dodd (1934), buying assets at this discount to their real value tends to generate extraordinary returns on a recurring basis. Later, Graham (1949) added the concept of margin of safety, a central element for the practical application of the philosophy.

Decades later, Warren Buffett became the main representative of this approach, achieving consistent returns of approximately 20% per year since 1965, when he took control of Berkshire Hathaway. Buffett also contributed to refining the strategy by emphasizing that the most important factor in choosing an asset is the company's sustainable competitive advantages. In this way, he relaxed the rules of his mentor, arguing that an exceptional company, even when trading at a relatively high price, can be a good investment.

Over the years, several researchers have tested the profitability of value and growth stocks to determine which strategy would be more advantageous in the long term. Fama and French (1992) showed that value stocks consistently outperform growth stocks in the US market between 1963 and 1990. Lakonishok, Shleifer and Vishny (1994) also reached the same result when comparing the return of the two stock classes.

2.4 Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is a financial model that describes the relationship between risk and the expected return of a financial asset (Sharpe, 1964). The CAPM starts from the premise that investors are risk-averse and seek to maximize their returns. It establishes that the expected return of a financial asset is determined by two components: the risk-free return and an additional risk premium. In the original article, Sharpe (1964) presented the basic concept of the CAPM, while Lintner (1965) and Mossin (1966) contributed to refining the model in subsequent works.

Gunnar (2016) applied the CAPM to portfolios formed from the regression between the portfolio returns and the Ibovespa, seeking to verify the validity of the EMH, according to which it is impossible to beat the market return without taking additional risk. The results were not significant enough to confirm this hypothesis. All portfolios obtained a statistically significant beta below 1, showing that they were less exposed to the systematic market risk. All alphas are positive, but only three of the eighteen values are statistically significant at the 90% confidence level.

Persson and Selander (2009) tested the CAPM and obtained similar results in the Nordic market. The regression betas are statistically significant, but the alphas were not significant. Zeidler (2014) and Milane (2016), applying the model in the Brazilian context, found results different from those of Gunnar (2016), with statistically significant betas and alphas for most of the portfolios analyzed.

2.5 Risk factors

Risk factors were first developed by Fama-French (1992) as a complement to the CAPM, adding the size and value factors to the market factor studied by Sharpe (1964). The model became known as the Fama-French 3 Factor Model, and Carhart (1997) later added the Momentum factor. Factors are used to describe the excess return of a portfolio, making it possible to assess the portfolio's exposure to different risk dimensions and to identify any alpha generated by active management.

Frazzini, Kabiller and Pedersen (2013) applied factor decomposition to Buffett's portfolio at Berkshire Hathaway. Under the CAPM, the portfolio showed an annual alpha of approximately 19% in the period from 1976 to 2011. However, when applying the Carhart (1997) model, the estimated alpha fell to about 12% per year, showing that part of the superior performance was associated with exposure to the value factor.

They then added the Betting-Against-Beta and Quality-Minus-Junk factors, and the portfolio's alpha fell to 6% per year, which indicated significant exposure to the quality factor as well.

This study adopts the Carhart (1997) model as the basis for measuring the alpha generated by the portfolios analyzed.

2.6 Modern Portfolio Theory

Modern Portfolio Theory (MPT), formulated by Markowitz (1952) in the seminal article titled *Portfolio Selection*, brought about a profound transformation in the approach to the investment process. This approach introduced the concepts of diversification and the balance between risk and return, proposing that investors can maximize the expected return of a portfolio for a given level of risk, or minimize risk for a specific level of desired return. The central premise is that investors can optimize their portfolios by combining different assets, taking into account the correlation between them.

Ljungberg and Högstedt (2021) applied MPT to Magic Formula portfolios to investigate whether returns could be improved by optimizing the portfolio allocation. In the book, Greenblatt suggests that investors allocate resources equally among the stocks chosen for the portfolio, disregarding the volatility and correlation between assets.

Ljungberg and Högstedt (2021) created three portfolios: (i) equal allocation, (ii) maximizing the Sharpe ratio and (iii) minimizing the portfolio variance. The results indicated that all portfolios beat the market, but those built according to MPT criteria showed higher returns and higher Sharpe ratios, reinforcing the hypothesis that careful asset selection combined with an optimized allocation can improve the results of an investment strategy.

3 METHODOLOGY

This section describes the methodology adopted in this work, detailing the research steps, which comprise data collection, portfolio construction, and the measurement and evaluation of results.

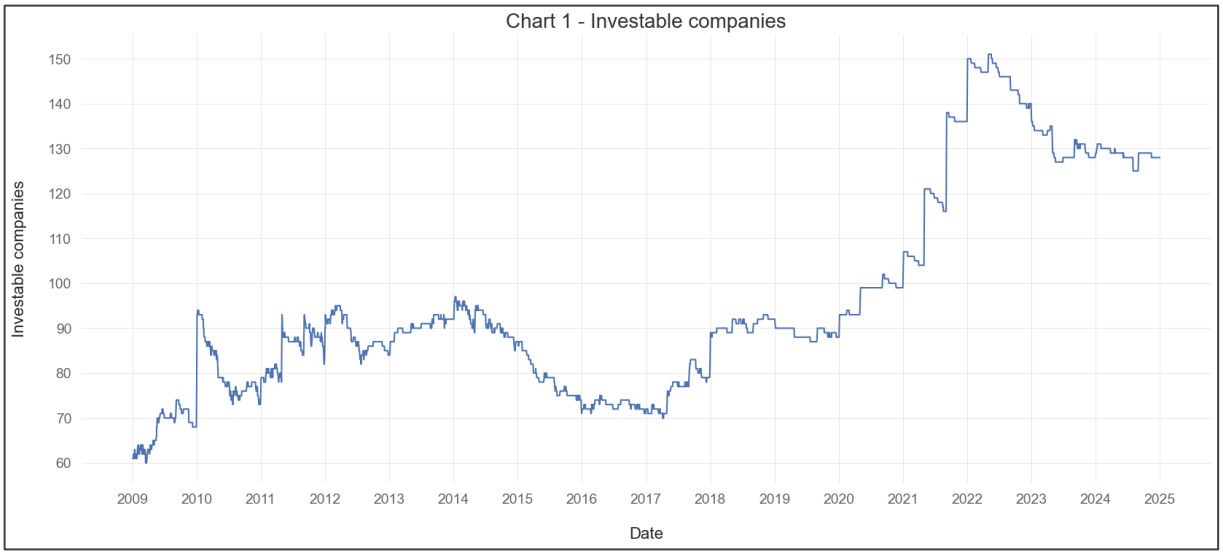
3.1 Data collection and filters

The data used in this study were obtained from the Comdinheiro platform, covering the historical composition of B3's Small Caps (SMLL) and Mid-Large Caps (MLCX) indices, selected to represent the universe of investable companies. The companies' accounting information and market prices were collected for the period between 2009 and 2024.

The filters applied to the initial universe followed the methodology proposed by Greenblatt (2006; 2010), which excludes companies with low liquidity, those with a market value below 10 million dollars, specific sectors such as Utilities and Financials (including banks, insurers, funds and brokers), as well as foreign companies represented by Brazilian Depositary Receipts (BDRs).

To adapt these restrictions to the context of the Brazilian market, the liquidity criterion was modified: (i) the median traded value over the last 42 days must be above one million reais; and (ii) if a company has multiple trading codes, the one with the highest median volume on the rebalancing date is selected. Zeidler (2014) argues that the market value filter originally used by Greenblatt is not fully applicable to the Brazilian market, given the difference in market structure.

In addition, the historical composition of the SMLL and MLCX indices was used to mitigate survivorship bias, taking into account events such as mergers, delistings and bankruptcies over the period analyzed. The number of stocks available for investment after applying the filters is shown in Chart 1.



Source: Prepared by the author. The chart shows the number of companies available for investment after applying the liquidity filter and eliminating companies in the Financial and Utilities sectors.

3.2 Asset ranking and portfolio construction

The asset ranking method presented by Greenblatt (2006) consists of:

1. Calculating the indicators: Return on Capital (ROC) and Earnings Yield (EY);
2. Sorting the stocks in descending order of ROC and EY, scoring each asset according to its position in the ranking. The result is two tables showing the stocks with the best ROC and the stocks with the best EY;
3. Adding up the asset's score in the two tables.
4. Sorting the assets' final score in ascending order to form the final ranking used in the asset selection.

This way, the portfolio will hold highly profitable and cheap companies. Companies without the data needed to calculate the indicators on the portfolio construction date and companies with negative EBIT were removed from the ranking.

3.2.1 Return on Capital

Greenblatt (2006) defines Return on Capital as:

$$ROC = \frac{EBIT}{Net\ Working\ Capital + Net\ Fixed\ Assets}$$

The formula consists of operating income before taxes and financial expenses, represented by Earnings Before Interest and Taxes (EBIT), divided by Net Working Capital (NWC) plus the company's "tangible" assets. The EBIT used corresponds to the accumulated value of the last 12 months, while the values of working capital and fixed assets are taken from the balance sheet of four quarters earlier, that is, the capital at the start of the period in which the EBIT was accumulated. The calculation method for NWC and Net Fixed Assets is described below:

$$NWC = Current\ Assets - (Current\ Liabilities - Short - term\ loans)$$

$$Net\ Fixed\ Assets = Total\ Assets - Current\ Assets - Intangibles$$

According to Greenblatt (2006), using EBIT instead of net income seeks to eliminate distortions arising from variations in leverage and taxation across companies and sectors, emphasizing the profit generated by the main operation. The term tangible assets, in the author's definition, refers to the capital effectively invested in the company's operation, unlike equity or total assets. This includes both the short-term capital needed to maintain operating activities and the capital invested in infrastructure, reflected in fixed assets.

3.2.2 Earnings Yield

Unlike the literature on the topic, Greenblatt (2006) defines Earnings Yield as:

$$EY = \frac{EBIT}{Market\ Value + Net\ Debt}$$

This metric was chosen because it covers both the portion belonging to shareholders and that belonging to creditors, reflecting the operating return generated relative to the total cost of buying the company as a whole. In addition, using EBIT in the numerator helps to eliminate distortions caused by different levels of leverage and taxation, as discussed in the previous section.

The EBIT considered corresponds to the accumulated value of the last 12 months. Market value is the sum of common and preferred shares multiplied by the share price on the ranking calculation date. Net debt is taken from the company's latest available balance sheet, also as of the ranking calculation date.

3.2.3 Portfolio construction

Greenblatt (2006) recommends building portfolios of 20 to 30 assets, in order to diversify non-systematic risk. According to the author, purchases should take place in lots of 5 to 7 assets every two or three months, and the assets should be held for 12 months, being sold at the end of this period to be replaced by new assets selected according to their position in the ranking.

However, much of the literature on the topic differs from the author's recommendations. Gunnar (2016) had difficulty trying to follow the methodology described by Greenblatt in the Brazilian market, as the same stocks kept appearing over short periods of 2 and 3 months. In view of this, the study built portfolios of 5, 10 and 15 assets, with semiannual and annual rebalancing. In addition, Gunnar isolated the factors of the formula to assess the individual impact of each component on portfolio performance.

Zeidler (2014) and Milane (2016) followed the same methodology as Olin (2011), but applied to the Brazilian market. Both built portfolios of 5, 10 and 15 assets and varied the holding periods between 3, 6, 9, 12, 15 and 18 months; all three authors built directional portfolios with the best and worst in the ranking to compare the results.

In contrast, Persson and Selander (2009) stayed closer to Greenblatt's original methodology, buying two assets per month to form a portfolio of 24 assets at the end of the first year. The assets were held for 12 months, being replaced monthly on a rolling basis.

All the studies mentioned above chose to distribute weights equally among the portfolio's stocks, as recommended by Greenblatt. Ljungberg and Högstedt (2021) allocated the

assets based on the Markowitz Efficient Frontier, creating one portfolio that optimizes return and another that minimizes risk.

In this study, we chose to build the following portfolios: (i) a portfolio of 24 assets, buying six assets every three months, following the methodology proposed by Greenblatt (2006); (ii) five monthly rebalanced portfolios, formed according to the companies' position in the ranking; and (iii) six Long-Short portfolios, which pit the best against the worst companies in the ranking, also rebalanced monthly. In all portfolios, weights were distributed equally among the assets.

3.3 Measurement and evaluation of results

This part describes the calculation methodology for the cumulative return, the portfolio risk and the Sharpe ratio.

3.3.1 Cumulative Return

The portfolio return is calculated every time the portfolio is rebalanced. It is described by the sum of the returns of each stock in the portfolio weighted by the weight of each one, as shown in the formula below:

$$r_c = \sum_{i=1}^{n} w_{i,t-1} * \left(\frac{V_{i,t}}{V_{i,t-1}} - 1 \right)$$

$$w_{i,t} = \frac{V_{i,t}}{V_{c,t}}$$

Where:

- r_c = Weighted average return of the portfolio;
- t = Rebalancing period;
- $w_{i,\,t}$ = Weight of asset i at t;
- $V_{i,\,t}$ = Value of asset i at t;
- $V_{c,\,t}$ = Total value of the portfolio;
- n = number of assets in the portfolio.

The total return for the period is calculated by compounding the returns of each period. After calculating the cumulative return, we calculated the average annual return of the portfolios to compare them with the Ibovespa and the CDI. This value is described as the compound annual growth rate (CAGR), where:

$$r_{t_o,t_i} = \left(\frac{V_{t_i}}{V_{t_0}} \right)^{\left(\frac{1}{t_n - t_0} \right)} - 1$$

Where:

- $r_{t0,\,ti}$ = average annual return or CAGR of the portfolio for period i;
- V_{ti} = Value of the portfolio at the end of period i;
- V_{t0} = Value of the portfolio at the start of the period;

- $t_n - t_0$ = difference between the first and the last period;

3.3.2 Sharpe Ratio

To measure the risk-adjusted return of the portfolios, we used the Sharpe Ratio, which quantifies the additional return obtained per unit of risk taken by the investor. This metric is widely used to compare the performance of different investments, helping in decisions related to capital allocation by allowing a more efficient balance between return and risk. The Sharpe Ratio was originally proposed by Sharpe (1966) and its formulation was refined in a later study (Sharpe, 1994); the author defines the number as:

$$S_c = \frac{r_c - r_f}{\sigma_c}$$

$$\sigma_{c,i} = \left(\sqrt{\frac{\sum_{n=1}^{n} (r_i - \bar{r})^2}{n-1}} \right)$$

Where:

- S_c = Sharpe ratio of the portfolio;
- r_c = Weighted average return of the portfolio;
- r_f = Average return of the risk-free rate;
- σ_c = Annualized volatility of the portfolio;
- $\sigma_{c,i}$ = Volatility of the portfolio in period i;
  For this analysis, the risk-free rate used was the CDI, as it is the reference rate of the Brazilian market.

3.3.3 Carhart Four-Factor Model

The decomposition of portfolio returns by risk factors is obtained from a multiple linear regression of the portfolio's monthly excess returns on the monthly returns of the factors. Excess return is measured as the return above the risk-free rate. The daily return of the factors was collected from the Center for Research in Financial Economics of the University of São Paulo (NEFIN-USP) and compounded into monthly returns. The relationship between the portfolio return and the factor returns is expressed as follows:

$$r_c - r_f = \alpha + \beta_1 MKT + \beta_2 SMB + \beta_3 HML + \beta_4 WML + \varepsilon$$

Where:

- r_c = Monthly return of the portfolio;
- r_f = Monthly return of the risk-free rate;
- α = Portfolio alpha, the return not explained by the factors;
- β_n = Exposure of the portfolio to the factor;
- ε = Random error;
- MKT = Market premium;

- SMB = Size factor or size premium;
- HML = Value factor or value premium;
- WML = Momentum factor.

To obtain more consistent results, the risk-free rate adopted was also provided by NEFIN-USP.

4 PRESENTATION OF RESULTS

This section presents the results of the portfolios described above. The Ibovespa was adopted as the benchmark for the directional (Long) portfolios, while the CDI was used for the Long-Short portfolios. We then decompose the results by risk factors in order to identify the portfolios' exposure. The results below do not take transaction costs into account.

4.1 Portfolio performance

The results of the directional strategies show that there is a relationship between the companies' position in the ranking and their historical performance, although not a strict one: the first quintile returned considerably more than the last, but the third returned more than the second. The portfolio built with the Magic Formula showed positive performance over the period analyzed, although below that of the first quintile, as shown in Table 1. The Magic Formula's CAGR was 12.0%, with a Sharpe of 0.12, while the first quintile showed a CAGR of 13.9% and a Sharpe of 0.19. The Ibovespa, over the same period, returned 7.0% per year with a Sharpe of -0.10. A similar performance was observed in the Long-Short strategies: the best result was obtained by the portfolio long the 25% best-ranked companies and short the 25% worst. According to Table 2, this strategy showed a CAGR of 3.3% and a Sharpe ratio of 0.21.

Table 1 – Cumulative return of the directional (Long) portfolios, 2009-2024

	1st Quintile	2nd Quintile	3rd Quintile	4th Quintile	5th Quintile	Magic Formula	Ibovespa
CAGR	13.9%	9.0%	11.2%	7.2%	7.1%	12.0%	7.0%
Volatility	23.6%	23.3%	24.2%	25.2%	28.6%	22.0%	24.1%
Sharpe	0.19	-0.02	0.07	-0.09	-0.08	0.12	-0.10

Source: Prepared by the author. The Sharpe of the directional portfolios is calculated as: Sharpe = (CAGR – CDI) / Volatility.

Table 2 – Cumulative return of the Long-Short (Best x Worst) portfolios, 2009-2024

	10%	20%	25%	30%	40%	50%	CDI
CAGR	1.3%	2.7%	3.3%	2.5%	1.5%	0.1%	9.4%
Volatility	25.5%	17.3%	15.6%	14.3%	12.2%	10.4%	-
Sharpe	0.05	0.16	0.21	0.17	0.12	0.01	-

Source: Prepared by the author. The Sharpe of the Long-Short portfolios is calculated as: Sharpe = CAGR / Volatility.

Volatility increases considerably in the portfolios with the worst companies, a difference of 5.0 percentage points per year in volatility between the first and the last quintile. In the Long-Short strategies, the portfolio volatility decreases as the number of companies on both legs increases. The portfolio of the 10% best holds on average 9.2 assets on the Long leg, while the portfolio of the 50% best holds 42.5 assets; the values are practically the same on the Short leg. This happens because, unlike quintiles, which were intervals within the sample universe, the

Long-Short portfolios select the ends of the sample and move towards the center as the percentile increases.

Table 3 – Number of stocks per portfolio

	Quintiles	10% LS	20% LS	25% LS	30% LS	40% LS	50% LS
Minimum	8	6	11	13	16	21	26
Maximum	28	14	28	34	41	54	67
Average	16.8	9.2	17.5	21.8	26.0	34.0	42.5

Source: Prepared by the author. The Long-Short columns show the Long leg; the minimum and the maximum of the Short leg differ by at most one stock.

The variation in the number of assets in the portfolios directly affects the return of each one. Quintiles are 20% slices of the universe of investable companies, so the number of companies in the portfolio changes as the number of investable companies increases, unlike the Magic Formula portfolio, which constantly holds 24 assets, except in the first year. It can thus be inferred that the performance of the Magic Formula portfolio is below the 1st Quintile because of the greater diversification of assets, since it holds stocks from both the 1st and the 2nd quintile for most of the period analyzed. Table 3 shows the minimum, maximum and average number of stocks for all portfolios built from percentiles. The portfolios divided into quintiles held an average of 16.8 assets.

4.2. Comparison with benchmarks

As seen in Table 1, the main directional strategies beat their benchmark. However, the average return does not explain all of a portfolio's performance. For it to be possible to develop a product from the Magic Formula, performance needs to be consistent over the period. Table 4 shows the annual performance of the main strategies seen above. The Magic Formula beat the Ibovespa in 10 of the 16 years in the sample, or 63% of the time, including 5 consecutive years, and the 1st Quintile in 8 of the 16 years, including 4 consecutive years. Returns were not always positive: the Magic Formula had 6 negative years, as many as the Ibovespa.

The performance of the directional strategies differs greatly from that of the Long-Short strategies: the directional portfolios suffered considerable losses in the crisis years between 2011 and 2015, in which the Long-Short ones lost less or gained, although they stayed above the CDI only in 2011. On the other hand, the Long portfolios performed very well in the years of recovery from the crises, when the Long-Short portfolios lagged behind them. Over the whole sample, the Long-Short portfolios in Table 4 stayed above the CDI in only 4 of the 16 years: 2011, 2017, 2021 and 2022.

Table 4 – Annual return between 2009 and 2024

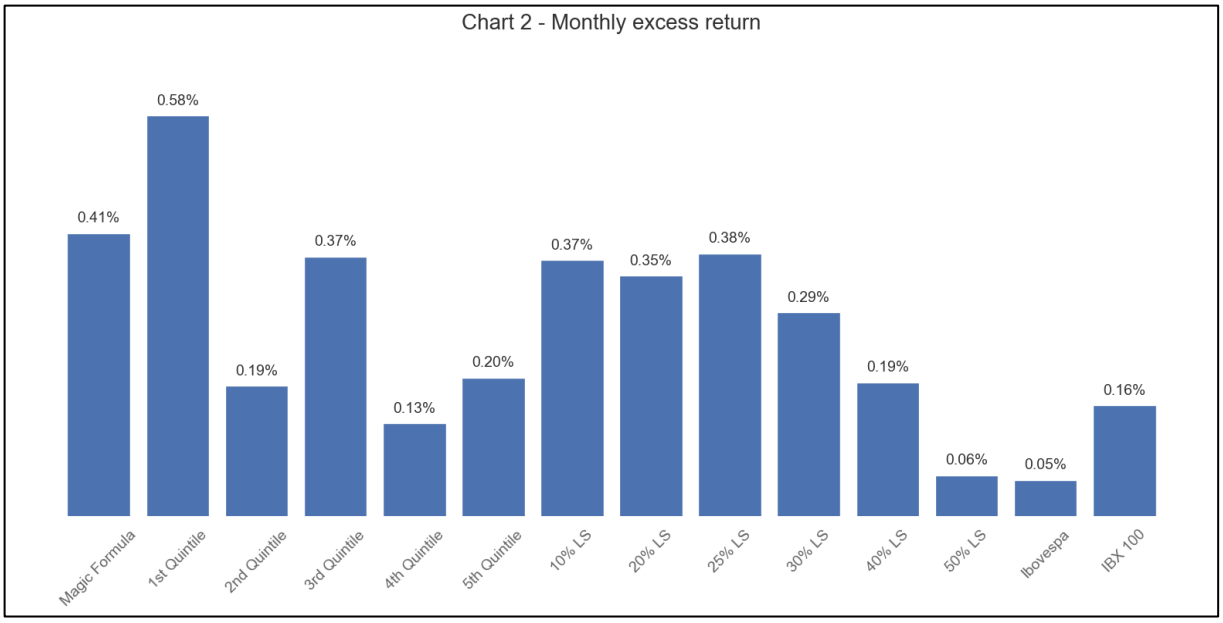
Year	1st Quintile	2nd Quintile	Magic Formula	Ibovespa	20% LS	25% LS	CDI
2009	68.7%	118.3%	82.1%	**70.4%**	-24.2%	-19.6%	**9.8%**
2010	17.8%	12.7%	18.5%	**1.0%**	5.4%	8.3%	**9.7%**
2011	-10.5%	-8.0%	-8.9%	**-18.1%**	16.3%	15.8%	**11.6%**
2012	39.8%	46.0%	38.4%	**7.4%**	0.7%	2.3%	**8.4%**
2013	-4.3%	-17.1%	-7.9%	**-15.5%**	5.4%	-3.4%	**8.1%**
2014	-12.5%	-8.0%	-12.9%	**-2.9%**	4.9%	5.1%	**10.8%**
2015	-28.9%	-2.5%	-26.8%	**-13.3%**	-5.9%	3.4%	**13.2%**
2016	71.4%	24.9%	48.1%	**38.9%**	-5.2%	3.6%	**14.0%**
2017	99.8%	30.2%	78.6%	**26.9%**	20.6%	14.4%	**10.0%**
2018	-1.1%	-0.2%	7.6%	**15.0%**	-11.7%	-15.2%	**6.4%**
2019	60.6%	35.7%	51.0%	**31.6%**	-6.8%	-5.8%	**6.0%**
2020	2.7%	0.1%	4.5%	**2.9%**	-6.7%	-6.8%	**2.8%**
2021	9.9%	-11.0%	3.6%	**-11.9%**	42.0%	36.6%	**4.4%**
2022	-2.0%	1.2%	-13.2%	**4.7%**	41.6%	46.8%	**12.4%**
2023	16.1%	17.1%	17.7%	**22.3%**	-10.7%	-13.6%	**13.0%**
2024	-22.6%	-30.7%	-19.8%	**-11.1%**	0.1%	0.7%	**10.8%**
Total	**700.0%**	**296.2%**	**510.2%**	**196.4%**	**53.6%**	**67.1%**	**322.1%**

Source: Prepared by the author. The Total row shows the cumulative return over the period.

4.3 Factor decomposition of performance

Applying Carhart's (1997) Four Factor Model assesses the alpha generated by the Magic Formula and the other strategies tested, identifying which risk factors contributed to their performance. The portfolios' exposure to these factors was estimated through a time-series linear regression, in which the monthly excess return was the dependent variable and the monthly factor premiums were the independent variables; alpha corresponds to the constant of the equation. Chart 2 shows that the Magic Formula generated a 0.41% monthly premium over the risk-free rate, while the Ibovespa generated a 0.05% premium above the risk-free rate over the period. Neither premium is statistically different from zero (p-values of 37.8% and 90.6%).

As shown in Table 5, the Magic Formula obtained an alpha of 0.22% per month, which is not statistically significant (p-value of 30.6%), and a market beta of 0.92. The portfolio has relevant exposure to SMB, with a beta of 0.32, and low exposure to WML and HML. The quintile portfolios behaved similarly to the Magic Formula: exposure to SMB increases as the companies' position in the ranking worsens, and exposure to MKT is highest in the two worst quintiles. This shows that the worst-ranked are mostly small companies that are riskier than the others.



Source: Prepared by the author. Excess return is defined as the portfolio return minus the risk-free rate, that is, the premium obtained for investing in the strategy. The risk-free rate used was provided by NEFIN-USP.

The Long-Short portfolios showed results opposite to those observed in the directional portfolios. As shown in Table 6, these strategies had negative exposure to MKT and SMB.

Interestingly, all portfolios showed low or neutral exposure to HML, and only 3 of the 12 coefficients in Tables 5 and 6 are statistically significant. Given that Greenblatt's methodology is grounded in Value Investing and adopts Earnings Yield as a selection criterion, one would expect higher exposure to this factor.

Table 5 – Factor decomposition of the directional portfolios (2009-2024)

	Magic Formula	1st Quintile	2nd Quintile	3rd Quintile	4th Quintile	5th Quintile
4-Factor Alpha	**0.22%**	**0.29%**	**0.03%**	**0.24%**	**-0.07%**	**0.05%**
	(0.21%)	(0.24%)	(0.24%)	(0.24%)	(0.26%)	(0.27%)
MKT	**0.92*****	**0.98*****	**0.92*****	**0.93*****	**1.10*****	**1.12*****
	(0.04)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
SMB	**0.32*****	**0.35*****	**0.38*****	**0.45*****	**0.45*****	**0.55*****
	(0.04)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
HML	**-0.03**	**-0.01**	**-0.14*****	**-0.12*****	**-0.15*****	**-0.01**
	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)	(0.05)
WML	**-0.04**	**0.01**	**-0.02**	**-0.06***	**-0.03**	**-0.14*****
	(0.03)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
Adjusted R^2	**80.9%**	**78.3%**	**75.6%**	**78.8%**	**79.0%**	**83.0%**

Source: Prepared by the author, * 10% statistical significance, ** 5% statistical significance, *** 1% statistical significance, standard errors in parentheses. Excess return is the difference between the portfolio's monthly return and the return of the risk-free rate over the same period. Alphas are the coefficients of the regressions based on each risk model, and factors are the betas relative to the historical return of each factor. The historical series of factor premiums and the risk-free rate were collected from NEFIN-USP.

Table 6 – Factor decomposition of the Long-Short portfolios (2009-2024)

	10% LS	20% LS	25% LS	30% LS	40% LS	50% LS
4-Factor Alpha	**0.16%**	**0.22%**	**0.30%**	**0.22%**	**0.14%**	**0.02%**
	(0.50%)	(0.35%)	(0.33%)	(0.30%)	(0.26%)	(0.23%)
MKT	**-0.09**	**-0.14****	**-0.19*****	**-0.18*****	**-0.16*****	**-0.10****
	(0.10)	(0.07)	(0.06)	(0.06)	(0.05)	(0.04)
SMB	**-0.23****	**-0.19*****	**-0.15****	**-0.11****	**-0.13****	**-0.10****
	(0.10)	(0.07)	(0.06)	(0.06)	(0.05)	(0.04)
HML	**-0.09**	**0.01**	**0.04**	**0.03**	**0.01**	**0.00**
	(0.09)	(0.06)	(0.06)	(0.05)	(0.05)	(0.04)
WML	**0.24*****	**0.14****	**0.10***	**0.09***	**0.08***	**0.06***
	(0.08)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)
Adjusted R^2	**12.1%**	**13.5%**	**11.6%**	**11.7%**	**13.8%**	**10.0%**

Source: Prepared by the author, * 10% statistical significance, ** 5% statistical significance, *** 1% statistical significance, standard errors in parentheses. Excess return is the difference between the portfolio's monthly return and the return of the risk-free rate over the same period. Alphas are the coefficients of the regressions based on each risk model, and factors are the betas relative to the historical return of each factor. The historical series of factor premiums and the risk-free rate were collected from NEFIN-USP. The Long-Short portfolios are self-financed, so their dependent variable is the return of the portfolio itself.

Table 7 compares the factor exposure of the main portfolios with the market indices. The three strategies based on Greenblatt's methodology showed higher alphas than the Ibovespa and the IBrX-100, whose alphas are negative and statistically significant. However, the alphas of the three strategies are not statistically significant, so the results do not establish abnormal returns above those of the benchmarks. The directional Magic Formula portfolios also showed higher exposure to SMB, because of the SMLL companies that are not part of the main indices, which increased the sensitivity to the factor.

Table 7 – Comparison with the benchmarks (2009-2024)

	Magic Formula	1st Quintile	20% LS	Ibovespa	IBX 100
4-Factor Alpha	**0.22%**	**0.29%**	**0.22%**	**-0.16%****	**-0.07%***
	(0.21%)	(0.24%)	(0.35%)	(0.07%)	(0.04%)
MKT	**0.92*****	**0.98*****	**-0.14****	**1.05*****	**1.04*****
	(0.04)	(0.05)	(0.07)	(0.01)	(0.01)
SMB	**0.32*****	**0.35*****	**-0.19*****	**0.01**	**-0.02*****
	(0.04)	(0.05)	(0.07)	(0.01)	(0.01)
HML	**-0.03**	**-0.01**	**0.01**	**0.05*****	**0.02*****
	(0.04)	(0.04)	(0.06)	(0.01)	(0.01)
WML	**-0.04**	**0.01**	**0.14****	**-0.03****	**0.00**
	(0.03)	(0.04)	(0.05)	(0.01)	(0.01)
Adjusted R^2	**80.9%**	**78.3%**	**13.5%**	**97.5%**	**99.2%**

Source: Prepared by the author, * 10% statistical significance, ** 5% statistical significance, *** 1% statistical significance, standard errors in parentheses. Excess return is the difference between the portfolio's monthly return and the return of the risk-free rate over the same period. Alphas are the coefficients of the regressions based on each risk model, and factors are the betas relative to the historical return of each factor. The historical series of factor premiums and the risk-free rate were collected from NEFIN-USP. The Long-Short portfolios are self-financed, so their dependent variable is the return of the portfolio itself.

5 CONCLUSION

The study shows that Greenblatt's Magic Formula beat the Ibovespa over the period analyzed, with a CAGR of 12.0% against 7.0%, although with negative years: 6 of 16, as many

as the index. These findings are in line with the results of Milane (2016) and Gunnar (2016), whose portfolios also beat the benchmark.

The quintile analysis showed that performance broadly follows the companies' position in the ranking: the first quintile returned considerably more than the last (13.9% against 7.1% per year), although the order among the quintiles is not perfect. Portfolios made up of the best-ranked companies delivered higher returns with lower volatility than the worst, indicating higher-quality and lower-risk assets. In addition, the Long-Short strategies showed a positive premium, the highest when buying the 25% best and selling the 25% worst, but without statistical significance: the alpha of that portfolio is 0.30% per month, with a p-value of 36.5%. Diversification also proved to be an important driver of performance: very concentrated or excessively diversified portfolios showed worse results. The results suggest that there is an optimal point of diversification at which the best risk-adjusted return can be extracted.

The factor decomposition, based on Carhart's (1997) Four-Factor Model, showed that the Magic Formula has a positive alpha that is not statistically significant (0.22% per month, with a p-value of 30.6%), and a lower exposure to MKT than the Ibovespa. Among the portfolios in this study, only two variations in the Appendix have alphas significant at the 5% level: the Magic Formula with Net Debt/Ebit and the 1st Quintile with Momentum. The decomposition result supports the results of the performance and risk analysis of the portfolios: the market beta is higher in portfolios with the worst companies in the ranking. SMB follows the same trend, showing that small companies are usually the worst in the ranking.

Greenblatt (2006) stresses that applying the strategy requires discipline and patience, recommending that it be maintained for at least three years to achieve the expected returns. In this study, the longer the holding period, the more often the Magic Formula portfolio beat the Ibovespa: it did so in 65% of the rolling twelve-month windows, 68% of the three-year windows and 92% of the five-year windows.

The results suggest that financial products based on the Magic Formula, such as funds or investment clubs, could be feasible, given the higher return than the Ibovespa with somewhat lower volatility (22.0% against 24.1%). However, the backtest does not include transaction costs, the maximum drawdown was deeper than that of the index (-55.1% against -48.6%) and the alpha is not statistically significant, so feasibility still depends on further analysis, especially of costs. The potential is lower for Long-Short strategies, since adding transaction costs to the backtest could eliminate the alpha observed in the study, especially the cost of stock lending.

The capital capacity of this strategy is limited by the liquidity of the Brazilian stock market. The filter adopted in this study — median daily volume above R$ 1 million over the last 2 months — is adequate for smaller operations but restrictive for large funds, which could distort prices when executing relevant orders. In that case, it would be necessary to split orders or adopt stricter liquidity filters, with a direct impact on the results.

This limitation, however, does not apply to individual investors, whose capital is smaller, allowing them to exploit less liquid assets without major restrictions. In addition, the methodological simplicity of the Magic Formula makes it accessible to unsophisticated investors.

6 FUTURE RESEARCH

This study sought to remain faithful to the methodology described by Greenblatt (2006), adapting it to the specificities of the Brazilian market. However, several aspects can still be explored in future research, some already investigated in other countries and which can be replicated in Brazil.

Ljungberg and Högstedt (2021), for example, built portfolios using Markowitz's (1952) Modern Portfolio Theory, with the aim of minimizing risk and maximizing the Sharpe ratio,

achieving better results than equal-weight allocation. Another possibility would be to assign weights based on the companies' market value. However, given the high concentration of the Brazilian stock market in a few stocks, such as Vale and Petrobras, this approach would reduce the diversification of the portfolios and make it harder to identify the source of the strategy's alpha.

In addition, assessing the operating costs of the directional and Long-Short strategies is a relevant step towards understanding their viability as a financial product. Quarterly rebalancing of only a quarter of the portfolio tends to reduce brokerage costs because of the low turnover. The greatest complexity lies in measuring stock lending costs for Long-Short strategies, given the low liquidity of the securities lending market.

Another promising line of research would be to combine the Magic Formula with other factors, such as Momentum, Low Volatility or Betting Against Beta, in order to avoid companies with short-term stressed prices. Given that part of the strategy involves buying discounted companies, there is a risk that this discount stems from persistent recent market declines, which could compromise the results over the investment horizon.

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APPENDIX

The following section explores variations of the Magic Formula by adding other factors to the stock ranking, as mentioned in Section 6. The first is Net Debt/Ebit, measured as Short-term debt + Long-term debt – Cash and equivalents, with Ebit being the accumulated value of the 4 most recent quarters; companies with Ebit below 0 or missing information for any of the quarters are discarded. The use of this indicator is based on the findings of Penman, Richardson and Tuna (2007) and George and Hwang (2009), who explored the effect of financial leverage on company performance; the studies showed that value companies with low leverage outperform value companies with high leverage and leveraged growth companies. The Momentum indicator is measured according to the methodology of Jegadeesh and Titman (1993), which considers the return of the last 12 months excluding the most recent one. Finally, the Low Volatility indicator measures the volatility of the last 12 months; Pim Van Vliet and Blitz (2007) studied the volatility effect and found that portfolios with less volatile stocks generate more return in the long term than more volatile stocks.

The portfolios below follow the same construction as the portfolios of the study: the first replicates Greenblatt's (2006) construction with the added signals, the second is divided into quintiles and the third is the Long-Short of the first and last quintile. The new portfolios were created from the indicators: (i) ROC + EY + Net Debt/Ebit, (ii) ROC + EY + Momentum, (iii) ROC + EY + Low Vol.

Table 8 – Cumulative return of the portfolios based on the Magic Formula, 2009-2024

	Magic Formula	MF + Net Debt/Ebit	MF+Momentum	MF+Low Vol	Ibovespa
CAGR	12.0%	14.8%	16.6%	10.7%	7.0%
Volatility	22.0%	21.9%	21.3%	19.4%	24.1%
Sharpe	0.12	0.25	0.34	0.07	-0.10

Source: Prepared by the author. The Sharpe of the directional portfolios is calculated as: Sharpe = (CAGR – CDI) / Volatility.

Table 9 – Cumulative return of the 1st Quintile of the portfolios, 2009-2024

	Magic Formula	MF + Net Debt/Ebit	MF+Momentum	MF+Low Vol	Ibovespa
CAGR	13.9%	13.7%	19.6%	8.6%	7.0%
Volatility	23.6%	22.8%	23.2%	19.4%	24.1%
Sharpe	0.19	0.19	0.44	-0.04	-0.10

Source: Prepared by the author. The Sharpe of the directional portfolios is calculated as: Sharpe = (CAGR – CDI) / Volatility.

Table 10 – Cumulative return of the Long-Short (1Q – 5Q) portfolios, 2009-2024

	Magic Formula	MF + Net Debt/Ebit	MF+Momentum	MF+Low Vol	CDI
CAGR	2.7%	3.4%	11.1%	-2.3%	9.4%
Volatility	17.3%	18.1%	20.7%	21.2%	-
Sharpe	0.16	0.19	0.54	-0.11	-

Source: Prepared by the author. The Sharpe of the Long-Short portfolios is calculated as: Sharpe = CAGR / Volatility.

Table 8 showed that adding the Leverage and Momentum factors improves the Magic Formula's result, reducing volatility and increasing return. In Table 9, only Momentum

improves the return of the 1st Quintile; the Leverage factor reduces volatility, but the return is slightly lower (13.7% against 13.9%). The Low Vol factor considerably reduced the volatility of the Magic Formula, as expected, but failed to improve the portfolio's return and had the worst Sharpe among the portfolios tested, except for the Ibovespa.

Table 10 shows the result of the Long-Short portfolios: adding the Leverage factor caused a marginal improvement over the Magic Formula, with a higher CAGR and Sharpe but increased volatility. The Momentum factor contributed positively to the Magic Formula, multiplying the portfolio's return by about four and its Sharpe by more than three. Finally, the Low Vol factor again delivered the worst performance and showed the highest volatility among the Long-Short portfolios.

Replicating the analysis of Table 7, the performance of the new portfolios was analyzed with the Carhart 4-Factor Model, seeking to understand the alpha generated by the portfolios over the period studied. The tables below show the contribution of the factors for each of the new versions of the Magic Formula: the MF Portfolio column shows the backtest based on Greenblatt's method, the 1st Quintile column refers to the portfolio of the best 20% in the sample, and the 20% LS column corresponds to the 1Q-5Q portfolio.

Table 11 – Factor exposure of the MF + Net Debt/EBIT portfolio (2009-2024)

	MF Portfolio	1st Quintile	20% LS	Ibovespa	IBX 100
4-Factor Alpha	**0.42%****	**0.34%**	**0.32%**	**-0.16%****	**-0.07%***
	(0.20%)	(0.23%)	(0.34%)	(0.07%)	(0.04%)
MKT	**0.90*****	**0.93*****	**-0.22*****	**1.05*****	**1.04*****
	(0.04)	(0.04)	(0.07)	(0.01)	(0.01)
SMB	**0.39*****	**0.36*****	**-0.15****	**0.01**	**-0.02*****
	(0.04)	(0.04)	(0.07)	(0.01)	(0.01)
HML	**-0.09****	**-0.09****	**-0.10***	**0.05*****	**0.02*****
	(0.04)	(0.04)	(0.06)	(0.01)	(0.01)
WML	**-0.02**	**-0.01**	**0.16*****	**-0.03****	**0.00**
	(0.03)	(0.04)	(0.05)	(0.01)	(0.01)
Adjusted R^2	**81.6%**	**78.8%**	**22.2%**	**97.5%**	**99.2%**

Source: Prepared by the author, * 10% statistical significance, ** 5% statistical significance, *** 1% statistical significance, standard errors in parentheses. Excess return is the difference between the portfolio's monthly return and the return of the risk-free rate over the same period. Alphas are the coefficients of the regressions based on each risk model, and factors are the betas relative to the historical return of each factor. The historical series of factor premiums and the risk-free rate were collected from NEFIN-USP. The Long-Short portfolios are self-financed, so their dependent variable is the return of the portfolio itself.

Table 12 – Factor exposure of the MF + Momentum portfolio (2009-2024)

	MF Portfolio	1st Quintile	20% LS	Ibovespa	IBX 100
4-Factor Alpha	**0.34%**	**0.49%****	**0.55%**	**-0.16%****	**-0.07%***
	(0.21%)	(0.23%)	(0.36%)	(0.07%)	(0.04%)
MKT	**0.95*****	**0.98*****	**-0.17****	**1.05*****	**1.04*****
	(0.04)	(0.04)	(0.07)	(0.01)	(0.01)
SMB	**0.34*****	**0.40*****	**-0.14****	**0.01**	**-0.02*****
	(0.04)	(0.04)	(0.07)	(0.01)	(0.01)
HML	**-0.08****	**-0.05**	**-0.05**	**0.05*****	**0.02*****
	(0.04)	(0.04)	(0.06)	(0.01)	(0.01)
WML	**0.13*****	**0.17*****	**0.46*****	**-0.03****	**0.00**
	(0.03)	(0.04)	(0.06)	(0.01)	(0.01)
Adjusted R^2	**80.0%**	**79.0%**	**39.9%**	**97.5%**	**99.2%**

Source: Prepared by the author, * 10% statistical significance, ** 5% statistical significance, *** 1% statistical significance, standard errors in parentheses. Excess return is the difference between the portfolio's monthly return and the return of the risk-free rate over the same period. Alphas are the coefficients of the regressions based on each risk model, and factors are the betas relative to the historical return of each factor. The historical series of factor premiums and the risk-free rate were collected from NEFIN-USP. The Long-Short portfolios are self-financed, so their dependent variable is the return of the portfolio itself.

Table 13 – Factor exposure of the MF + Low Vol portfolio (2009-2024)

	MF Portfolio	1st Quintile	20% LS	Ibovespa	IBX 100
4-Factor Alpha	**0.10%**	**-0.05%**	**0.09%**	**-0.16%****	**-0.07%***
	(0.17%)	(0.19%)	(0.34%)	(0.07%)	(0.04%)
MKT	**0.77*****	**0.81*****	**-0.46*****	**1.05*****	**1.04*****
	(0.03)	(0.04)	(0.07)	(0.01)	(0.01)
SMB	**0.24*****	**0.17*****	**-0.44*****	**0.01**	**-0.02*****
	(0.03)	(0.04)	(0.07)	(0.01)	(0.01)
HML	**-0.08****	**-0.10*****	**-0.14****	**0.05*****	**0.02*****
	(0.03)	(0.03)	(0.06)	(0.01)	(0.01)
WML	**-0.02**	**-0.02**	**0.14****	**-0.03****	**0.00**
	(0.03)	(0.03)	(0.05)	(0.01)	(0.01)
Adjusted R^2	**80.6%**	**77.4%**	**49.3%**	**97.5%**	**99.2%**

Source: Prepared by the author, * 10% statistical significance, ** 5% statistical significance, *** 1% statistical significance, standard errors in parentheses. Excess return is the difference between the portfolio's monthly return and the return of the risk-free rate over the same period. Alphas are the coefficients of the regressions based on each risk model, and factors are the betas relative to the historical return of each factor. The historical series of factor premiums and the risk-free rate were collected from NEFIN-USP. The Long-Short portfolios are self-financed, so their dependent variable is the return of the portfolio itself.

The Magic Formula portfolio combined with the leverage factor, in Table 11, showed better results than the original portfolio, with an alpha of 0.42% per month, significant at the 5% level, against 0.22% without significance in the original portfolio. The portfolio combined with Momentum, in Table 12, showed a lower alpha than the leverage factor (0.34%, with a p-value of 10.6%), even with the highest CAGR among the 24-stock portfolios. Looking at the exposure of the two portfolios, the Momentum portfolio has more exposure to WML and MKT, which implies a reduction in alpha. In the 1st Quintile the order is reversed: Momentum has the highest alpha (0.49%, significant at the 5% level), and that of the leverage factor is not significant (0.34%). Table 13 shows the performance of the Magic Formula combined with the Low Vol factor, which again had the worst alpha generation among all tests; a relevant point in this portfolio is its lower exposure to MKT than the other portfolios, a common feature of minimum variance portfolios.

In line with the original Magic Formula portfolio, all Long-Short portfolios showed positive alpha, but none was statistically significant. As expected, adding the Momentum factor considerably increased this portfolio's exposure to WML. The Low Vol portfolio showed high negative exposure to the MKT and SMB factors, showing that this portfolio tends to buy larger companies with a lower market beta; even with statistical significance, the other portfolios do not have high exposure to these two factors.

Finally, the study showed that adding the leverage and Momentum indicators can improve the performance of the Magic Formula created by Joel Greenblatt, generating more alpha, slightly reducing the strategy's volatility and maintaining an exposure to the risk factors of the Carhart 4-Factor Model similar to that of the original portfolio, apart from the higher WML of the Momentum portfolio; the Low Vol indicator reduces volatility, but also alpha. These results should be read with care: the two alphas significant at the 5% level have p-values of 4.0% and 3.3%, no adjustment was made for the number of portfolios tested, and the significance of the Magic Formula with Net Debt/Ebit fell to the 10% level in variations measured during the review, such as keeping real estate companies in the universe or applying only part of the corrections listed in the note on this edition.